

❖ **Renewal Processes**

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Motivation.

A **renewal process** models repeated events over time where **waiting times between events are i.i.d.**

Think real world systems:

- machine repairs,
- component replacements,
- customer arrivals with general interarrival times,
- biological regeneration cycles,
- experiments restart after random durations.

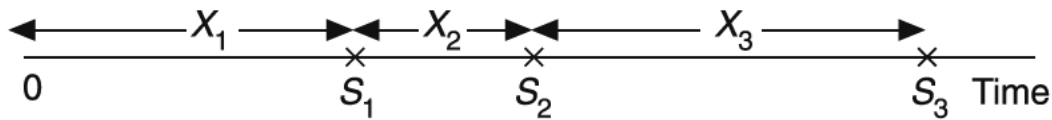
It generalizes the **Poisson process** by removing the exponential assumption.

Each cycle “restarts the system”, which is what **renewal** means.

Interarrival Times X_n : The Building Blocks

Let $\{X_n\}_{n \geq 1}$ be **(i.i.d.)** random variables with:

- $X_n > 0$ (time must move forward),
- Distribution CDF $F(x)$
- mean $\mu = E[X_1]$ (possibly infinite).



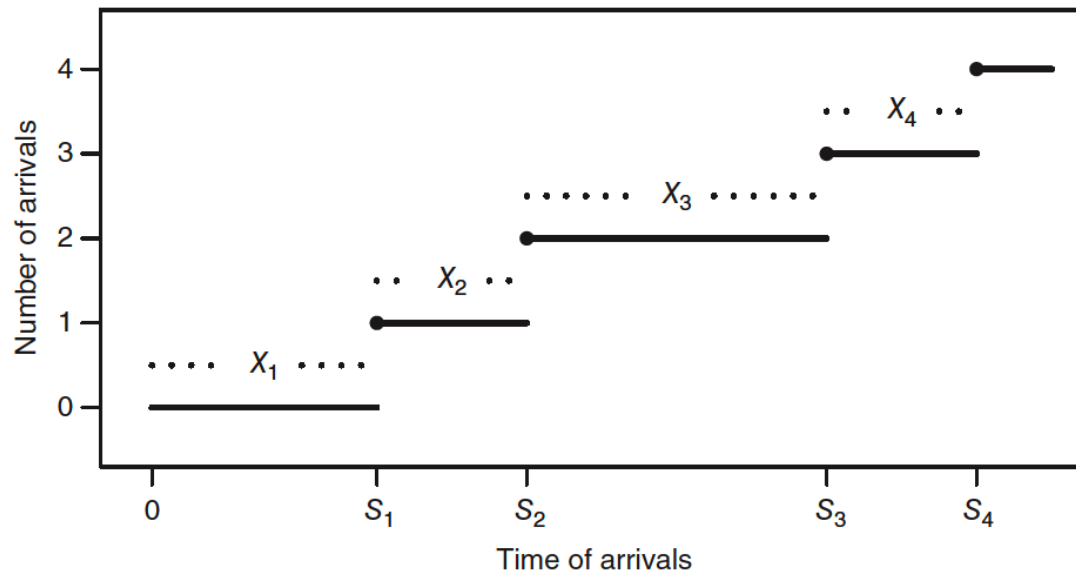
This is *weaker* than the Poisson process, where iid exponential gives memorylessness.

Arrival Times S_n :When Renewals Occur

The time of the n -th renewal is defined as the cumulative sums:

$$S_n = X_1 + X_2 + \cdots + X_n \text{ and } S_0 = 0$$

$\{S_n\}$ is a random walk with positive increments, but interpreted along the **time axis**.



Counting Process $N(t)$:Inverting Time

The **number of renewals** have happened by time t ,

$$N(t) = \max\{n : S_n \leq t\}$$

Equivalently,

$$N(t) = \sum_{n=1}^{\infty} \mathbb{I}(S_n \leq t)$$

- Arrival times are easier to define than counts.
- In applications, we observe counts over time.

Remarks

- Renewal Processes Are NOT Markov.
- Knowing $N(t) = k$ does **not** tell us how long we have waited since the last renewal.
- The age since the last renewal also matter.
- If X_n is exponential the age becomes irrelevant.
- Poisson is the only Markov renewal process.

Distribution of $N(t)$

The distribution of $N(t)$ can be obtained by

$$\mathbb{P}(N(t) = n) = \mathbb{P}(S_n \leq t < S_{n+1})$$

since the number of renewals by time t is greater than or equal to n if and only if the n -th renewal occurs before or at time t . ($N(t) \geq n \iff S_n \leq t$)

Sum of iid distributions is given by convolution, for $W = X + Y$

$$f_W(w) = \int_{-\infty}^{\infty} f_X(x) f_Y(w - x) dx$$

Distribution of $S_n = X_1 + X_2 + \cdots + X_n$ is given by n convolutions, which have no closed formula.

□ Renewal Equation (Key Structural Result)

Instead of full distribution, we study the mean behavior.

Renewal Function: $m(t) := E[N(t)] = \sum_{n=1}^{\infty} \mathbb{P}(S_n \leq t) = \sum_{n=1}^{\infty} \mathbb{P}(N(t) \geq n)$

Renewal Equation

$$m(t) = F(t) + \int_0^t m(t-x) dF(x)$$

This is a recursive self-similarity: after the first renewal, the process restarts.

Proof:

$$m(t) = E[N_t]$$

$$= E[E[N_t|X_1]]$$

$$= \int_0^\infty E[N_t|X_1 = x] f_X(x) dx$$

$$= \int_0^\infty \mathbb{I}_{\{t \geq s\}} (1 + E[N_{t-s}]) f_X(x) dx$$

$$= \int_0^t (1 + m(t-s)) f_X(x) dx$$

$$= F(t) + \int_0^t m(t-x) dF(x)$$

Elementary Renewal Theorem

If $\mu = E[X_1] \leq \infty$, then

$$\lim_{t \rightarrow \infty} \frac{m(t)}{t} = \frac{1}{\mu}$$

Meaning:

- long-run average renewal rate depends **only on the mean**
- not on variance, skewness, or shape of F

Renewal theory is LLN in continuous time.

LLN: $\frac{S_n}{n} \rightarrow \mu$

Almost Sure Law for $N(t)$

Theorem

$$\frac{N(t)}{t} \xrightarrow{a.s.} \frac{1}{\mu}$$

Proof idea:

$$S_{N(t)} \leq t \leq S_{N(t)+1}$$

Use LLN on S_n and sandwich.

Central Limit Theorem for Renewal Process:

Let $\mu = E[X]$ and $\sigma^2 = Var[X]$, and define

$$Z(t) = \frac{N(t) - t/\mu}{\sqrt{t\sigma^2/\mu^3}}$$

Then, $Z(t)$ converges to a standard normal Z as $t \rightarrow \infty$, that is for any a we have

$$\lim_{t \rightarrow \infty} \mathbb{P}(Z(t) \leq a) = \mathbb{P}(Z \leq a) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^a e^{-\frac{x^2}{2}} dx$$

Further more,

$$t^{-1}Var[N(t)] \rightarrow \frac{\sigma^2}{\mu^3}$$

- **Renewal Reward Processes**

Consider a taxi driver who drops off passengers at times t_n , $n \geq 1$, where $\{t_n\}$ forms a renewal process with i.i.d. interarrival times:

$$X_n = t_n - t_{n-1}$$

Let R_j denote the fare paid by the j -th passenger.

We interpret R_j as a **reward** earned by the driver.

We assume that the time spent searching for the next passenger is negligible, so that a new passenger is found immediately after the previous one is dropped off.

Goal: Compute the **long-run rate at which the driver earns money**, (dollars per unit time.)

Renewal Reward Theorem

At each renewal: Earn reward **iid** R_n (profit, uptime, cost, etc.)

Total reward by time t :

$$R(t) = \sum_{n=1}^{N(t)} R_n$$

Assumptions: Assume that the pairs $\{(X_n, R_n)\}_{n \geq 1}$ are i.i.d. This allows the reward R_j to depend on the corresponding ride length X_j , but not on any other interarrival times or rewards.

Renewal Reward Theorem

$$\frac{R(t)}{t} \xrightarrow{a.s.} \frac{E[R_1]}{E[X_1]}$$

Example (Taxi Driver Revisited: Including Passenger Search Time)

We now refine the taxi driver model by incorporating the fact that the driver must spend time **searching for new passengers**.

Assume

- Y_j denote the amount of time spent finding the j -th passenger after the $(j - 1)$ -st passenger departs,
- L_j denote the length of the j -th passenger's ride,
- R_j denote the fare paid for the j -th ride.

Define $X_j = L_j + Y_j$ as a cycle for the driver earning a total rewards R_j .

The Renewal Reward Theorem applies: the long-run rate at which the driver earns money is

$$\frac{E[R_1]}{E[X_1]} = \frac{E[R_1]}{E[L_1] + E[Y_1]}$$

Example (Train Dispatching Problem)

Suppose that passengers arrive at a train platform according to a renewal process with rate μ . As soon as N passengers have arrived, a train departs carrying all N passengers. This process repeats indefinitely.

Assume the following cost structure for the train company:

- When exactly n passengers are waiting on the platform, the company incurs a cost at rate nc dollars per unit time.
- Each train departure incurs a fixed cost of K dollars.

Our goal is to compute the **long-run average cost rate** incurred by the train company.

Renewal structure

We take **train departures** as the renewal epochs and define one cycle as the time between consecutive train departures.

- s_n denote the arrival time of the n -th passenger,
- $T_n = s_n - s_{n-1}$ denote the interarrival times of passengers.

We assume $\{T_n\}$ are i.i.d. with $E[T_i] = \frac{1}{\mu}$

A cycle ends once N passengers have arrived. Therefore, the **cycle length** is

$$X = T_1 + T_2 + \cdots + T_N,$$

and

$$E[X] = NE[T] = \frac{N}{\mu}$$

Cost accumulated during a cycle

$$R = 0 \cdot cT_1 + 1 \cdot cT_2 + \cdots + (N - 1)cT_N + K.$$

Taking expectations,

$$\begin{aligned} E[R] &= cE[T](1 + 2 + \cdots + (N - 1)) + K \\ &= c \frac{1}{\mu} \cdot \frac{N(N - 1)}{2} + K \end{aligned}$$

By the **Renewal Reward Theorem**, the long-run average cost per unit time is

$$g(N) = \frac{E[R]}{E[X]} = \frac{c \frac{1}{\mu} \cdot \frac{N(N - 1)}{2} + K}{N/\mu} = \frac{c(N - 1)}{2} + \frac{K\mu}{N}$$

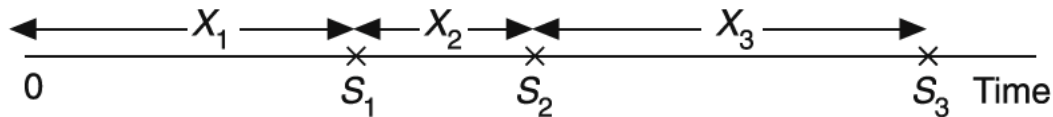
average **waiting cost**, increasing in N + average **dispatch cost per unit time**, decreasing in N

Optimization: For fixed numerical numbers c, μ, K , we can find N to optimize $g(N)$.

Example. *average age*

For this example we can view the renewal process as counting the number of times a machine breaks down. Each breakdown is independent, and occurs after a random time X . The times between breakdowns are IID, so this is a renewal process. The '**age**' of the device is

$$A(t) = t - S_{N(t)} = \text{time since last breakdown}$$



We then define the reward earned up to time t as

$$R(t) = \int_0^t A(s) ds$$

so the average age at time t is

$$\text{Average age} = \frac{R(t)}{t}$$

The graph of $A(t)$ is a sawtooth with drops to zero at each arrival time. The reward earned between times T_{k-1} and T_k is

$$R_k = \text{area under graph} = \frac{1}{2} X_k^2$$

By the renewal theorem we have

$$\text{Average age} = \frac{R(t)}{t} \rightarrow \frac{E[R]}{E[X]} = \frac{\frac{1}{2} E[X^2]}{E[X]}$$

For example if X is exponential with rate λ (so a Poisson process) then

$$\text{Average age} \rightarrow \lambda^{-1}$$

Example: The Insurance Ruin Problem.

❖ References:

Recommended textbooks

- Introduction to Probability Models - Sheldon M. Ross (Chapter 7)
- Probability and Random Processes - Geoffrey Grimmett and David Stirzaker
- Introduction to Probability, Statistics, and Random Processes - Hossein Poshro-Nik
- Introduction to Stochastic Processes with R - Robert Dobrow

More textbooks:

- Essentials of Stochastic Processes – Richard Durrett
- Understanding Markov Chains- Nicolas Privault
- Markov Chains –J.R. Norris
- Stochastic Processes, an Introduction. –Peter Jones and Peter Smith
- Stochastic Processes 2nd Edition by Sheldon M. Ross